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Parenting stressors, social support, and mental health of mothers of children with attention deficit hyperactivity disorder

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ABSTRACT

Background: Mothers of children with attention deficit hyperactivity disorder (ADHD) are known to have a tendency toward poor mental health. This study aimed to identify specific parenting stressors impacting maternal mental health as well as the social support that buffers against those stressors in mothers of children with ADHD.
Methods: A cross-sectional survey was conducted with Japanese mothers of children under 18 years of age diagnosed with ADHD. The Questionnaires included attributes of participants and their children with ADHD, parenting stressors assessed by the Developmental Disorder Parenting Stressor Index, social support provided from spouse, parents, peer, friends, school, and medical and welfare institution, and maternal mental health assessed by the K6. A total of 149 responses (mean age 43.1 ± 4.9, children's mean age 10.2 ± 3.3, with 80.5% male) were analyzed.
Results: Multivariate logistic regression analysis revealed that the parenting stressor 'difficulties in understanding the child and coping with the child's needs' was associated with poor maternal mental health. Additionally, social support from medical and welfare institution buffered the impact of this parenting stressor on maternal mental health.
Implications to practice: Medical and welfare institution professionals should strength social support reducing parenting stressors of 'difficulties in understanding the child and coping with the child's needs' among mothers of children with ADHD.
Conclusion: The parenting stressor of 'difficulties in understanding the child and coping with the child's needs' may impact mental health in mothers of children with ADHD and the medical and welfare institution's social support may buffer this stressor.
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Introduction

Mothers of children with ADHD often experience depressive symptoms. This has been supported by a meta-analysis showing that mothers of children with ADHD had significantly more severe depressive symptoms than other mothers (Cheung, Aberdeen, Ward and Theule, 2018). Such poor maternal mental health leads to dysfunctional parenting such as corporal punishment, lack of discipline, and overly strict disciplining (Gau & Chang, 2013; Gerdes, Haack and Schneider, 2012; Shin & Stein, 2008), worsening children's ADHD symptoms, behavioral problems, and oppositional defiant disorders (Agha, Zammit, Thapar and Langley, 2017; Brammer, Galán, Mesri and Lee, 2018; Chronis et al., 2007; Walls et al., 2018). Therefore, maintaining the mental health of mothers is important for both mothers and children with ADHD, and identifying factors associated with maternal mental health is crucial for developing effective measures.

Regarding factors related to the mental health of mothers of children with ADHD, a few studies have reported maternal attributes such as lower income (Al-Balushi et al., 2019), marital status (Chou, Hsiao, Chang and Yen, 2021), no other caregivers (Al-Balushi et al., 2019) and own ADHD tendencies (Marumoto et al., 2023). Additionally, children's related factors such as severity of ADHD symptoms (Cheung, Aberdeen, Ward and Theule, 2018), hyperactive/impulsive feature (Al-Balushi et al., 2019) and combined feature (Al-Balushi et al., 2019) have also been reported.

On the other hand, there has not been examined the relationship between parenting stressors and mental health of mothers of children with ADHD. Although qualitative studies exploring the parenting stressors experienced by mothers of children with ADHD remain limited (Leitch et al., 2019), previous research has identified several unique parenting stressors, including the difficulty in obtaining understanding from others (Laugesen & Groenkjaer, 2015; Leitch et al., 2019;

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Mofokeng & van der Wath, 2017) and the emotional struggle to accept their child's disability (Teraoka et al., 2014). There have been several assessment tool of parenting stressor to date. However, the widely used Parenting Stress Index (Abidin, 1983; Abidin, 1990; Abidin, 1995) does not clearly distinguish between parenting stressors and stress responses. In addition, although a stressor scale was developed by Song et al. (2004), and another scale for assessing parental stressor prior to kindergarten entry and school enrollment was developed by Watanabe et al. (2006), both scales were designed for parents of children with high-functioning autism or Asperger's disorder, and their validity has not been examined. On the other hand, The Developmental Disorder Parenting Stressor Index (DDPSI; Yamane, 2013) was designed to assess parenting stressors among mothers with developmental disabilities including ADHD and was demonstrated both reliability and validity.

According to DDPSI, the parenting stressors consist of four factors, 'difficulties in understanding the child and coping with the child's needs,' 'anxiety about the child's future and independence,' 'inadequate understanding of the child's disorder from others,' and 'conflicting emotions with regard to the child's disorder.' The highest proportion of variance in DDPSI was explained by the 'difficulties in understanding the child and coping with the child's needs.' However, it is unknown which of these parenting stressors has a negative effect on mental health of mothers of children with ADHD. A study has examined the association between parenting stressors and depressive symptoms in mothers of children with ADHD, but this used the Parenting Stress Index (PSI; Abidin, 1983; Abidin, 1990; Abidin, 1995) for mothers of children in general (Theule et al., 2013), moreover some items of PSI overlap those of depression measurement such as the Beck Depression Inventory (Beck, Ward, Mendelson, Mock and Erbaugh, 1961). Therefore, a study that identifies specific parenting stressors impacting mental health of mothers of children with ADHD is needed.

In addition, it has not been identified which provider's social support buffers the negative effects of parenting stressors on mental health of mothers of children with ADHD. In the context of stress and social support theory, a buffering effect refers to the role of social support in reducing the negative impact of stressors on individuals' mental health (Cohen & Wills, 1985). Social support, broadly defined as material and psychological assistance provided by others, is generally be divided into two types: instrumental support and emotional support (Vaux, 1988). In addition to the types of social support, previous research has reported that the effectiveness of social support differs depending on the providers of support (Acoba, 2024; Dakof & Taylor, 1990; Gariépy et al., 2016). In the present study, we focused on the providers of social support to determine which provider's social support are helpful. To date, it has been known that the total social support from husbands, parents, health care providers and peer parents is negatively related to parenting stressors in mothers of children with developmental disabilities (Yamane, 2013) and the total social support is negatively related to the total parenting stressors in parents of children with ADHD (Theule et al., 2011). It has also been known that social support is negatively associated with depressive symptoms in parents of children with ADHD (Lovell, Moss and Wetherell, 2012) and less family support is associated with depressive symptoms in caregivers of children with ADHD (Chou, Hsiao, Chang and Yen, 2021). Intervention studies have suggested that support from medical and welfare institution may contribute to improved maternal mental health. For example, behavioral parent training has been shown to improve child-related outcomes and to enhance mental health in parents of children with ADHD (Mehri et al., 2020a; Mehri et al., 2020b). However, it is unclear which provider's social support buffers the negative effects of parenting stressors on mental health of mothers of children with ADHD.

In Japan, various social supports are provided for parents of children with developmental disabilities. For example, developmental disability support centers, which are welfare institutions, mainly provide support for individuals with developmental disabilities, including ADHD, and their families (Ministry of Health, Labour and Welfare, 2021). In medical institutions, 'guidance for individual parents' to discuss how to deal with their children's troubling behaviors and 'behavioral parent training' are often conducted for parents of children with ADHD (Makino et al., 2018) based on the guidelines for diagnosis and treatment of the disorder (Saito & Iida, 2022). In schools, individual guidance is provided based on an understanding of the needs of each child with ADHD (Ministry of Education, Culture, Sports, Science and Technology, 2003), and special support education coordinators are assigned for this purpose (Ministry of Education, Culture, Sports, Science and Technology, 2018). Therefore, it is vital to identify the social support that buffers the negative effects of parenting stressors on mental health of mothers of children with ADHD.

Hence, this study aims to identify parenting stressors that affect the mental health of mothers of children with ADHD as well as providers of social support that buffer these effects. We hypothesize that the parenting stressor of 'difficulties in understanding the child and coping with the child's needs' is a risk factor for psychological distress and that social support provided by spouses and parents buffer parenting stressors.

Methods

Participants

Participants were recruited through two peer groups for parents of children with developmental disabilities, a provider of after-school care services, and a support group for children with developmental disabilities in Japan. Inclusion criteria was to be a Japanese mother of children under 18 years of age who had been diagnosed with ADHD by medical professionals. Of 164 study participants, 149 who provided complete responses to the survey and met the eligibility criteria (mean age 43.1 ± 4.9 years, mean age of children with ADHD 10.2 ± 3.3 years, 80.5% male) were included in the analysis.

Procedures

Online survey was conducted between September 24 and October 19, 2021. Before beginning the survey, participants were presented with an online informed consent form that explained the purpose of the study, the voluntary nature of participation, and data confidentiality. Only those who provided consent electronically were allowed to proceed to the questionnaire. This study was approved by the Research Ethics Committee of the Faculty of Health and Sport Sciences of the University of Tsukuba, Japan (approval number: Tai 021–8; approval date: June 19, 2021). This study was reported in accordance with the STROBE statement (von Elm et al., 2007).

Measures

Participant attributes

The respondents were asked about their age, education, marital status, occupation, household income, ADHD tendencies, number of children, number of children with ADHD, and psychiatric medication status.

To measure maternal ADHD tendencies, Part A of the Adult ADHD Self-Report Scale (ASRS; Kessler et al., 2005) was administered. The ASRS has two parts: Part A and Part B. Part A is used to screen adults for ADHD. Threshold-level responses to four or more of the six Part A questions indicate a positive result. The sensitivity and specificity of the Part A screening using the ASRS were 68.7% and 99.5%, respectively (Kessler et al., 2005). Cronbach's alpha coefficient for Part A of the ASRS in this study was 0.831. Mothers with a positive ASRS Part A result were considered as having ADHD tendencies.

Regarding psychiatric medication status, data on the mothers' medicine use for mental health problems (antidepressants, anxiolytics and hypnotics) and ADHD symptoms (ADHD medications) were collected. General drug names for antidepressants, anxiolytics, hypnotics, and

ADHD medications approved in Japan were provided as references in the questionnaire. The use of any of these medications was itemized as 'use of any psychiatric medications.'

Attributes of participants' children with ADHD

The attributes of the children with ADHD included age, sex, diagnosis of comorbid developmental disorders (autism spectrum disorders, learning disorders, developmental coordination disorders, and others), and ADHD symptom severity. In case of having two or more children with ADHD, participants were required to respond with regard to the child with the highest burden. ADHD symptom severity was assessed using the Japanese version of home form of the ADHD-Rating Scale (ADHD-RS; Tani et al., 2010). A higher score indicates greater severity of ADHD symptoms. Cronbach's alpha coefficient of ADHD-RS in this study was 0.878.

Parenting stressors

We used the DDPSI, which was specifically developed to assess parenting stressors that are universally experienced by mothers of children with developmental disabilities, including ADHD, regardless of the child's age (Yamane, 2013). The18-item questionnaire comprises four factors and respondents were asked to rate the frequency of their experience for each item using a 4-point Likert scale: (1) difficulties in understanding the child and coping with the child's needs (six items, score range: 0–18), (2) anxiety about the child's future and independence (five items, score range: 0–15), (3) inadequate understanding of the child's disorder from others (four items, score range: 0–12), and (4) conflicting emotions with regard to the child's disorder (three items, score range: 0–9). A higher score on any factor indicates greater stressor experienced by the respondents. The Cronbach's alpha coefficients of the four factors in this study ranged from 0.721 to 0.868.

Social support

The Social Support Scale (Yamane, 2013) was used with some modifications. It comprises four items for instrumental and emotional support, such as 'He/She is willing to talk to me about my child and discuss it with me,' 'He/She listens to my worries and complaints about parenting,' 'He/She cheers me up,' and 'He/She provides guidance, education, treatment, and assistance that is appropriate for my child.' Providers of social support were established based on previous studies as 'Spouse,' 'Parents (biological parent, parent-in-law),' 'Peer (a person met through a parents' meeting),' 'Friends (other than a person met through a parents' meeting),' 'School (such as a teacher and a special education coordinator),' and 'Medical and welfare institution (such as a doctor and a psychologist).' For each item, respondents were asked to rank the degree of expected support on a 4-point Likert scale (definitely no/probably no/probably yes/definitely yes). Score range of each social support is 4 to 16 and a higher score indicates a state in which respondents feel they can expect more support. The results of the principal component analysis for each of the six providers' support indicated that all had high loadings (0.80–0.93) and high contribution rates (72.96–83.76) for the first principal component, and all were confirmed to have a one-dimensional structure. The Cronbach's alpha coefficient for our respondents was 0.915 for the scale as a whole and 0.874–0.935 for the six provider's support.

Maternal mental health

We used the Japanese version of the K6, whose validity and reliability have been verified (Furukawa et al., 2008). The K6 is a scale designed to be sensitive around the threshold for the clinically significant range of the distribution of nonspecific distress (Kessler et al., 2003). Poor mental health was defined as a K6 score ≥ 9, which is one of the optimal cut-off points corresponding to DSM-IV (Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition) mood disorders and anxiety disorders and is typically adopted for the Japanese population (Dhungel et al., 2021; Hamazaki et al., 2016; Miyaki et al., 2012; Takaki & Hibino, 2014; Takehara et al., 2020). The Cronbach's alpha coefficient for K6 in this study was 0.866.

Statistical analysis

First, the relationships between maternal mental health and participant attributes, attributes of participants' children with ADHD, parenting stressors, and social support were examined using univariate logistic regression analysis with mental health (K6 score ≤ 8/K6 score ≥ 9) as the dependent variable. *P*-value correction applying the Benjamini-Hochberg method was conducted for multiple test problems. Second, multivariate logistic regression analysis (forced entry method) was conducted with mental health as the dependent variable and variables related to mental health in univariate analysis as independent variables.

Then, six multivariate logistic regression analyses were conducted with mental health as the dependent variable and the parenting stressors that had a significant association with mental health in the above multivariate logistic regression analysis (forced entry method) and each of six provider's social support as independent variables. In these six multivariate logistic regression analyses, participant attributes, attributes of participants' children with ADHD and parenting stressors which showed significant association with mental health in the univariate analysis were adjusted.

In addition to the primary multivariate logistic regression analyses, we conducted sensitivity analyses to examine the robustness of our findings. Specifically, for both the initial multivariate logistic regression analysis and the six subsequent multivariate models, we constructed additional models that included clinically relevant potential confounders— maternal ADHD tendencies, use of any psychiatric medications, and comorbid developmental disorders—regardless of their statistical significance in the univariate analyses.

In the analyses, parenting stressors and social support were divided into two categories by median split because none of their ten score distributions followed a normal distribution, as determined by the Kolmogorov–Smirnov test (Supplementary Fig. S1). Before the multivariate logistic regression analysis, we examined the interrelationships between the independent variables and confirmed that there was no multicollinearity. IBM SPSS Statistics version 27 for Macintosh was used for all statistical analyses, and reported values were considered statistically significant at $p < .05$.

Results

Mean K6 score of all respondents was 8.5 ± 5.3, and 74 (49.7%) respondents had poor mental health (score ≥ 9). Table 1 shows the result of logistic regression analysis for mental health. In the univariate analysis, mental health showed significant association with parenting stressors of difficulties in understanding the child and coping with the child's needs ($p < .001$, odds ratio = 5.02), anxiety about the child's future and independence ($p = .012$, odds ratio = 2.92), inadequate understanding of the child's disorder from others ($p = .006$, odds ratio = 3.31), conflicting emotions with regard to the child's disorder ($p = .014$, odds ratio = 2.76), use of any psychiatric medications ($p = .015$, odds ratio = 5.19) and severity of their child's ADHD symptoms ($p = .015$, odds ratio = 1.05).

In the multivariate analysis (forced entry method), parenting stressors of difficulties in understanding the child and coping with the child's needs ($p = .039$, odds ratio = 2.53) and use of any psychiatric medications ($p = .005$, odds ratio = 4.94) had a significant association with mental health. The overall logistic regression model was statistically significant ($\chi^2(6) = 40.06$, $p < .001$), indicating that the predictors collectively contributed to the model.

The six multivariate logistic regression analyses are presented in Table 2. Although all forms of social support had no independent relationship with mental health, only Model 6, with the inclusion of social

Table 1
Logistic regression analysis for maternal mental health.

		Mean ± SD or *n* (%)		Univariate analysis				Multivariate analysis			
				Odds ratio	95% CI		p^a	Odds ratio	95% CI		p
Parenting stressors											
Difficulties in understanding the child and coping with the child's needs	≤8 (Ref)	73	(49.0)								
	≥9	76	(51.0)	5.02	2.51	10.07	**<0.001**	2.53	1.05	6.11	**0.039**
Anxiety about the child's future and independence	≤11 (Ref)	74	(49.7)								
	≥12	75	(50.3)	2.92	1.50	5.69	**0.012**	1.89	0.88	4.04	0.101
Inadequate understanding of the child's disorder from others	≤4 (Ref)	70	(47.0)								
	≥5	79	(53.0)	3.31	1.69	6.48	**0.006**	1.96	0.88	4.37	0.102
Conflicting emotions with regard to the child's disorder	≤5 (Ref)	79	(53.0)								
	≥6	70	(47.0)	2.76	1.42	5.37	**0.014**	1.68	0.75	3.75	0.210
Social support											
Spouse	≤10 (Ref)	71	(47.7)								
	≥11	78	(52.3)	0.48	0.25	0.92	0.092				
Parents	≤10 (Ref)	67	(45.0)								
	≥11	82	(55.0)	0.60	0.31	1.15	0.278				
Peer	≤12 (Ref)	87	(58.4)								
	≥13	62	(41.6)	0.82	0.43	1.58	0.605				
Friends	≤11 (Ref)	80	(53.7)								
	≥12	69	(46.3)	0.70	0.37	1.34	0.434				
School	≤11 (Ref)	70	(47.0)								
	≥12	79	(53.0)	0.71	0.37	1.35	0.415				
Medical and welfare institution	≤13 (Ref)	74	(49.7)								
	≥14	75	(50.3)	1.35	0.71	2.56	0.470				
Attributes											
Age		43.1	±4.9	0.95	0.89	1.01	0.279				
Education	Without university education (Ref)	84	(56.4)								
	With university education	65	(43.6)	1.21	0.63	2.31	0.596				
Marital status	Married (Ref)	132	(88.6)								
	Unmarried	17	(11.4)	1.52	0.55	4.23	0.514				
Occupation	Employed (Ref)	105	(70.5)								
	Unemployed	44	(29.5)	0.76	0.37	1.54	0.507				
Household income	Less than 6 million (Ref)	66	(44.3)								
	More than 6.01 million	83	(55.7)	1.51	0.79	2.90	0.350				
ADHD tendencies	No (Ref)	112	(75.2)								
	Yes	37	(24.8)	2.29	1.06	4.95	0.101				
Number of children	Less than 2 (Ref)	118	(79.2)								
	3 or more	31	(20.8)	1.10	0.50	2.44	0.807				
Number of children with ADHD	1 (Ref)	114	(76.5)								
	2 or more	35	(23.5)	1.73	0.80	3.73	0.316				
Use of any psychiatric medications	No (Ref)	124	(83.2)								
	Yes	25	(16.8)	5.19	1.83	14.70	**0.015**	4.94	1.63	14.95	**0.005**
Attributes of participants' children with ADHD											
Age		10.2	±3.3	0.94	0.85	1.04	0.372				
Sex	Male (Ref)	120	(80.5)								
	Female	29	(19.5)	1.87	0.81	4.30	0.293				
Comorbid developmental disorders	Yes (Ref)	120	(80.5)								
	No	29	(19.5)	0.66	0.29	1.50	0.436				
Severity of ADHD symptoms		26.0	±10.2	1.05	1.02	1.09	**0.015**	1.16	0.52	2.58	0.726

Note. *SD*, Standard deviation; *CI*, confidence interval; *ADHD*, attention deficit hyperactivity disorder.
In multivariate analysis (forced entry method), variables related to mental health in univariate analysis such as 'use of any psychiatric medications,' 'child's severity of ADHD symptoms,' parenting stressors of 'difficulties in understanding the child and coping with the child's needs,' parenting stressors of 'anxiety about the child's future and independence,' parenting stressors of 'inadequate understanding of the child's disorder from others,' and parenting stressors of 'conflicting emotions with regard to the child's disorder' were used as independent variables.
[a] Adjusted *p*-value by Benjamini-Hochberg method.

support from medical and welfare institution, showed the vanishing of the significant association between the 'difficulties in understanding the child and coping with the child's needs' stressor and mental health ($p = .061$, odds ratio = 2.35). The results of sensitivity analyses after adjusting for potential confounders were also consistent with the main findings (Supplementary Tables S1 and S2), indicating that the observed associations remained robust.

Discussion

We aimed to identify parenting stressors which have an impact on maternal mental health as well as social support which buffer these effects in mothers of children with ADHD.

First, this study revealed that among parenting stressors measured by DDPSI, 'difficulties in understanding the child and coping with the child's needs' was associated with maternal mental health. This may be consistent with the fact that this factor explains the highest proportion of the variance in the DDPSI (Yamane, 2013). The factor 'difficulties in understanding the child and coping with the child's needs' was composed of questionnaire items such as 'I did not really know what my child was thinking,' 'It was difficult for me to understand the needs and desires of my child,' and 'I did not know what to do for my child when they caused problems.' This supports our hypotheses and suggests that reducing the parenting stressor of 'difficulties in understanding the child and coping with the child's needs' is vital to preventing mental health problems in mothers of children with ADHD. At the

Table 2
Social Support's Buffer Against the Effect of 'Difficulties in Understanding the Child and Coping with the Child's Needs' Stressor on Maternal Mental Health (Six Multivariate Logistic Regression Analyses).

			Model 1		Model 2		Model 3		Model 4		Model 5		Model 6	
	Level	*n*	Odds ratio (95% CI)	*p*	Odds ratio (95% CI)	*p*	Odds ratio (95% CI)	*p*	Odds ratio (95% CI)	*p*	Odds ratio (95% CI)	*p*	Odds ratio (95% CI)	*p*
Parenting Stressor														
Difficulties in understanding the child and coping with the child's needs	≤8	73	Ref		Ref		Ref		Ref		Ref		Ref	
	≥9	76	2.69 (1.09–6.59)	**0.031**	2.51 (1.03–6.10)	**0.042**	2.58 (1.06–6.26)	**0.036**	2.53 (1.05–6.11)	**0.040**	2.53 (1.05–6.11)	**0.039**	2.35 (0.96–5.73)	0.061
Social Support														
Spouse	≤10	71	Ref											
	≥11	78	0.55 (0.26–1.19)	0.128										
Parents	≤10	67			Ref									
	≥11	82			0.95 (0.44–2.03)	0.886								
Peer	≤12	87					Ref							
	≥13	62					0.79 (0.37–1.67)	0.531						
Friends	≤11	80							Ref					
	≥12	69							0.92 (0.43–1.98)	0.840				
School	≤11	70									Ref			
	≥12	79									1.04 (0.47–2.26)	0.932		
Medical and welfare institution	≤13	74											Ref	
	≥14	75											1.82 (0.83–4.00)	0.134

Note. *CI*, confidence interval.
In each analysis, variables related to mental health in univariate analysis such as 'use of any psychiatric medications,' 'child's severity of ADHD symptoms,' parenting stressors of 'anxiety about the child's future and independence,' parenting stressors of 'inadequate understanding of the child's disorder from others,' and parenting stressors of 'conflicting emotions with regard to the child's disorder' were adjusted.

same time, it should be noted that the relationship between the parenting stressor and maternal mental health may be bidirectional. While our findings suggest that the parenting stressor contributes to poor mental health, it is also possible that maternal psychological distress exacerbates perceptions of parenting difficulties. In addition to parenting stressors, use of any psychiatric medications showed a strong independent association with poor maternal mental health in the multivariate analysis (odds ratio = 4.94). This variable likely reflects a complex clinical context. On one hand, it may serve as a marker of more severe pre-existing or concurrent mental health conditions requiring pharmacological treatment. On the other hand, psychiatric medication use may also be a consequence of unmanaged parenting stress. Given the cross-sectional design of the study, the causal relationships among parenting stressors, psychiatric medication use, and maternal mental health cannot be determined. Nonetheless, the magnitude of this association highlights the importance of considering psychiatric medication use as a key covariate when examining maternal mental health among mothers of children with ADHD.

Second, contrary to previous studies (Chou, Hsiao, Chang and Yen, 2021; Yamane, 2013), this study found no direct association between any social support and mental health, but it revealed that only social support from medical and welfare institution buffers the negative impact of the parenting stressor 'difficulties in understanding the child and coping with the child's needs' on mental health. It should be noted that although a previous intervention study demonstrated improvements in parental mental health through behavioral parent training (Mehri et al., 2020a), the present study clarified that such support appeared to function as a factor buffering the association between the parenting stressor and maternal mental health. The difference between our findings and previous studies should be interpreted with methodological and conceptual differences. Prior studies demonstrating improvements in parental mental health were based on randomized controlled trial designs that evaluated the effects of clearly defined and structured behavioral parent training interventions. In contrast, the present study was conducted in a naturalistic setting using an online survey. Moreover, social support from medical and welfare institution in our study was not limited to behavioral parent training but included broader forms of support such as instrumental support and emotional support.

It is well known that pediatricians, psychologists, and staffs of welfare support facilities provide behavioral parent training, which is an educational approach of teaching parenting skills to parents of children with ADHD (Evans, Owens and Bunford, 2014). In Japan, behavioral parent training is the primary social support offered by medical and welfare institution (Makino et al., 2018; Ministry of Health, Labour and Welfare, 2021). Behavioral parent training has been reported to improve parenting behaviors and mental health in parents of children with ADHD (Mehri et al., 2020a; Rimestad et al., 2019). It should be noted that the change in odds ratio was small (from 2.53 to 2.35), and the shift in *p*-value was also marginal (from 0.039 to 0.061). Therefore, further research is needed to validate the findings of the buffering effect of social support provided by medical and welfare institution. In addition, from a clinical perspective, future research should explore the differential impact of various types of support. The social support scale used in this study included both instrumental support (e.g., providing guidance, education, treatment, and assistance tailored to the child's needs) and emotional support (e.g., listening to concerns and offering empathy regarding parenting). However, the scale was not originally designed to distinguish between these two types of support in analysis. Further studies are needed to examine which type of support is more effective in buffering parenting stressors in clinical settings.

Strengths and limitations

This study has several strengths. First, to our knowledge, this is the first study to identify the specific parenting stressor associated with

poor mental health as well as the social support that buffers against parenting stressor in mothers of children with ADHD. Furthermore, beyond clinical settings, these findings may also inform public health and policy strategies. By identifying which parenting stressor affects maternal mental health and which providers' social support are effective in buffering these effects, this study provides evidence to guide the development of parent-focused interventions and the prioritization of support resources. Importantly, such distinctions have not been clearly addressed in previous research. These insights may contribute to the development of more targeted and effective intervention strategies, as well as resource allocation efforts aimed at promoting maternal well-being among mothers of children with ADHD.

However, this study has several limitations. First, it was a cross-sectional study, and we could not prove causal relationships between mental health, parenting stressors, and social support. Thus, future longitudinal studies are required. Second, the sample size for this study was relatively small. Considering the possibility that the voluntary nature of the online survey and its distribution via support groups may have led to selection bias, we cannot rule out that the respondents are not representative of the population; specifically, these mothers have already reached to support groups, potentially leading to an underestimation of stress and mental health issues in the broader population. Third, in the analyses in this study, parenting stressors and social support were divided into two categories by median split because the distribution of both was bimodal with peaks at both ends of the median. However, this median split has not been clinically validated, there are interpretation limitations for a high vs low comparison based on this sample-based values. Specifically, dichotomizing continuous variables may reduce statistical power, obscure variability, and limit the interpretability of the results. Therefore, the findings should be interpreted with caution. Alternative approaches using standardized continuous scores or quantile regression should be considered in future research to preserve statistical power and information. Fourth, the survey used a self-report questionnaire that may have introduced reporting and recall biases. Further studies using objective measurements such as expert ratings of clinical symptoms would be useful. Fifth, it should be noted that our study is limited to Japanese mothers and may not be generalizable to other countries and contexts. Sixth, although the children in our sample spanned a wide age range, we were unable to conduct detailed analyses stratified by child age due to the limited sample size. Future studies with larger samples are needed.

Implications to practice

The findings of this study have several implications for clinical practice. First, the results suggest that the parenting stressor of 'difficulty in understanding the child and coping with the child's needs' may play a key role in poor maternal mental health among mothers of children with ADHD. Therefore, clinical support for these mothers should not only focus on managing the child's symptoms but also address this parenting stressor.

Second, the present findings suggest that social support from medical and welfare institution may buffer the negative impact of the parenting stressor on maternal mental health. In clinical settings, pediatricians working at medical institutions in departments such as child psychiatry, and clinical psychologists working at developmental disability support centers may play an important role in providing such support. In addition to offering behavioral parent training and practical guidance for managing children's behaviors, professionals in medical and welfare institutions should also provide emotional support to mothers of children with ADHD by offering reassurance and encouragement and by listening to their concerns and difficulties related to child rearing.

Furthermore, the social support scale used in this study included both instrumental support, such as providing guidance, education, treatment, and assistance tailored to the child's needs, and emotional support, such as listening to concerns and offering empathy regarding parenting. These findings suggest that both types of support may be important components of clinical care for families of children with ADHD. Although the present study was not designed to distinguish their relative contributions, integrating both practical guidance and empathic communication into routine clinical interactions may help improve maternal poor mental health.

Conclusion

This study showed that the stressor of difficulties in understanding the child and coping with the child's needs was associated with poor maternal mental health and that social support from medical and welfare institution buffered the impact of this parenting stressor on maternal mental health. These findings will contribute to developing measures to improve mental health in mothers of children with ADHD.

Authors contribution

Tatsuro Marumoto: Conceptualization, Methodology, Validation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Takafumi Monma:** Validation, Formal analysis, Writing – review & editing. **Yukinori Sawae:** Data curation, Writing – review & editing. **Fumi Takeda:** Conceptualization, Methodology, Validation, Formal analysis, Writing – review & editing, Project administration, Supervision.

Data statement

The data are not available due to privacy and ethical restrictions.

CRediT authorship contribution statement

Tatsuro Marumoto: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Takafumi Monma:** Writing – review & editing, Validation, Formal analysis. **Yukinori Sawae:** Writing – review & editing, Data curation. **Fumi Takeda:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Ethics approval

This study was approved by the Research Ethics Committee of the Faculty of Health and Sport Sciences of the University of Tsukuba, Japan (approval number: Tai 021–8; approval date: June 19, 2021).

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Declaration of competing interest

Tatsuro Marumoto used to be an employee of Eli Lilly Japan K.K., Takeda Pharmaceutical Company Limited and Janssen Pharmaceutical K.K., and is an employee of Nippon Boehringer Ingelheim Co., Ltd.. The other authors have no competing interests to declare.

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Appendix A. Supplementary data

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References

Abidin, R. R. (1983). *Parenting Stress Index: Manual, Administration Booklet, [and] Research Update*. Charlottesville, VA: Pediatric Psychology Press.

Abidin, R. R. (1990). *Parenting Stress Index-Short Form*. Charlottesville, VA: Pediatric Psychology Press.

Abidin, R. R. (1995). *Parenting Stress Index: Manual* (3rd ed.). Odessa, FL: Psychological Assessment Resources.

Acoba, E. F. (2024). Social support and mental health: The mediating role of perceived stress. *Frontiers in Psychology*, *15* Article 1330720. https://doi.org/10.3389/fpsyg.2024.1330720.

Agha, S. S., Zammit, S., Thapar, A., & Langley, K. (2017). Maternal psychopathology and offspring clinical outcome: A four-year follow-up of boys with ADHD. *European Child & Adolescent Psychiatry*, *26*(2), 253–262. https://doi.org/10.1007/s00787-016-0873-y.

Al-Balushi, N., Al Shekaili, M., Al-Alawi, M., Al-Balushi, M., Panchatcharam, S. M., & Al-Adawi, S. (2019). Prevalence and predictors of depressive symptoms among caregivers of children with attention-deficit/hyperactivity disorder attending a tertiary care facility: A cross-sectional analytical study from Muscat, Oman. *Early Child Development and Care*, *189*(9), 1515–1524. https://doi.org/10.1080/03004430.2017.1394850.

Beck, A. T., Ward, C. H., Mendelson, M., Mock, J., & Erbaugh, J. (1961). An inventory for measuring depression. *Archives of General Psychiatry*, *4*, 561–571. https://doi.org/10.1001/archpsyc.1961.01710120031004.

Brammer, W. A., Galán, C. A., Mesri, B., & Lee, S. S. (2018). Parental ADHD and depression: Time-varying prediction of offspring externalizing psychopathology. *Journal of Clinical Child and Adolescent Psychology*, *47*(sup1), S137–S149. https://doi.org/10.1080/15374416.2016.1183495.

Cheung, K., Aberdeen, K., Ward, M. A., & Theule, J. (2018). Maternal depression in families of children with ADHD: A meta-analysis. *Journal of Child and Family Studies*, *27*(4), 1015–1028. https://doi.org/10.1007/s10826-018-1017-4.

Chou, W. J., Hsiao, R. C., Chang, C. C., & Yen, C. F. (2021). Predictors of depressive symptoms in caregivers of children with attention-deficit/hyperactivity disorder: A one-year follow-up study. *International Journal of Environmental Research and Public Health*, *18*(16), 8835. https://doi.org/10.3390/ijerph18168835.

Chronis, A. M., Lahey, B. B., Pelham, W. E., Jr., Williams, S. H., Baumann, B. L., Kipp, H., ... Rathouz, P. J. (2007). Maternal depression and early positive parenting predict future conduct problems in young children with attention-deficit/hyperactivity disorder. *Developmental Psychology*, *43*(1), 70–82. https://doi.org/10.1037/0012-1649.43.1.70.

Cohen, S., & Wills, T. A. (1985). Stress, social support, and the buffering hypothesis. *Psychological Bulletin*, *98*(2), 310–357.

Dakof, G. A., & Taylor, S. E. (1990). Victims' perceptions of social support: What is helpful from whom? *Journal of Personality and Social Psychology*, *58*(1), 80–89. https://doi.org/10.1037//0022-3514.58.1.80.

Dhungel, B., Tsuguhiko, K., Ochi, M., Gilmour, S., Kachi, Y., & Takehara, K. (2021). Association of child's disability status with father's health outcomes in Japan. *SSM - Population Health*, *16* Article 100951. https://doi.org/10.1016/j.ssmph.2021.100951.

Evans, S. W., Owens, J. S., & Bunford, N. (2014). Evidence-based psychosocial treatments for children and adolescents with attention-deficit/hyperactivity disorder. *Journal of Clinical Child and Adolescent Psychology*, *43*(4), 527–551. https://doi.org/10.1080/15374416.2013.850700.

Furukawa, T. A., Kawakami, N., Saitoh, M., Ono, Y., Nakane, Y., Nakamura, Y., ... Kikkawa, T. (2008). The performance of the Japanese version of the K6 and K10 in the World Mental Health Survey Japan. *International Journal of Methods in Psychiatric Research*, *17*(3), 152–158. https://doi.org/10.1002/mpr.257.

Gariépy, G., Honkaniemi, H., & Quesnel-Vallée, A. (2016). Social support and protection from depression: Systematic review of current findings in Western countries. *The British Journal of Psychiatry: the Journal of Mental Science*, *209*(4), 284–293. https://doi.org/10.1192/bjp.bp.115.169094.

Gau, S. S., & Chang, J. P. (2013). Maternal parenting styles and mother-child relationship among adolescents with and without persistent attention-deficit/hyperactivity disorder. *Research in Developmental Disabilities*, *34*(5), 1581–1594. https://doi.org/10.1016/j.ridd.2013.02.002.

Gerdes, A. C., Haack, L. M., & Schneider, B. W. (2012). Parental functioning in families of children with ADHD: Evidence for behavioral parent training and importance of clinically meaningful change. *Journal of Attention Disorders*, *16*(2), 147–156. https://doi.org/10.1177/1087054710381482.

Hamazaki, K., Harauma, A., Otaka, Y., Moriguchi, T., & Inadera, H. (2016). Serum n-3 polyunsaturated fatty acids and psychological distress in early pregnancy: Adjunct study of Japan environment and children's study. *Translational Psychiatry*, *6*(2), e737. https://doi.org/10.1038/tp.2016.2.

Kessler, R. C., Adler, L., Ames, M., Demler, O., Faraone, S., Hiripi, E., ... Walters, E. E. (2005). The World Health Organization Adult ADHD Self-Report Scale (ASRS): A short screening scale for use in the general population. *Psychological Medicine*, *35*(2), 245–256. https://doi.org/10.1017/s0033291704002892.

Kessler, R. C., Barker, P. R., Colpe, L. J., Epstein, J. F., Gfroerer, J. C., Hiripi, E., ... Zaslavsky, A. M. (2003). Screening for serious mental illness in the general population. *Archives of General Psychiatry*, *60*(2), 184–189. https://doi.org/10.1001/archpsyc.60.2.184.

Laugesen, B., & Groenkjaer, M. (2015). Parenting experiences of living with a child with attention deficit hyperactivity disorder: A systematic review of qualitative evidence. *JBI Database of Systematic Reviews and Implementation Reports*, *13*(11), 169–234. https://doi.org/10.11124/jbisrir-2015-2449.

Leitch, S., Sciberras, E., Post, B., Gerner, B., Rinehart, N., Nicholson, J. M., & Evans, S. (2019). Experience of stress in parents of children with ADHD: A qualitative study. *International Journal of Qualitative Studies on Health and Well-Being*, *14*(1) Article 1690091. https://doi.org/10.1080/17482631.2019.1690091.

Lovell, B., Moss, M., & Wetherell, M. A. (2012). With a little help from my friends: Psychological, endocrine and health corollaries of social support in parental caregivers of children with autism or ADHD. *Research in Developmental Disabilities*, *33*(2), 682–687. https://doi.org/10.1016/j.ridd.2011.11.014.

Makino, K., Saito, K., Aoshima, M., Ito, C., Iwadare, Y., Kodaira, M., & Usami, M. (2018). Expert consensus on the diagnosis and treatment of ADHD in children: Psychosocial treatment and support edition. *Japanese Journal of Psychiatric Treatment*, *33*(9), 1113–1122.

Marumoto, T., Monma, T., Sawae, Y., & Takeda, F. (2023). Mental health and psychosocial status in mothers of children with attention deficit hyperactivity disorder (ADHD): Differences by Maternal ADHD Tendencies : Mental Health and Psychosocial Status in Mothers of ADHD Children. *Child Psychiatry and Human Development*, *56*(1), 34–42. https://doi.org/10.1007/s10578-023-01532-x.

Mehri, M., Chehrzad, M. M., Maleki, M., Kousha, M., Akhlaghi, E., & Mardani, A. (2020a). The effect of behavioral parent training of children with attention deficit hyperactivity disorder on parents' mental health. *Neurology Psychiatry and Brain Research*, *37*, 53–59. https://doi.org/10.1016/j.npbr.2020.06.003.

Mehri, M., Chehrzad, M. M., Mardani, A., Maleki, M., Dianatinasab, M., Kousha, M., & Assari, S. (2020b). The effect of behavioral parent training on sleep problems of school-age children with ADHD: A parallel randomized controlled trial. *Archives of Psychiatric Nursing*, *34*(4), 261–267. https://doi.org/10.1016/j.apnu.2020.04.001.

Ministry of Education, Culture, Sports, Science and Technology (2003). Key points on the future of special needs education (Final Report). Retrieved from https://www.mext.go.jp/b_menu/shingi/chousa/shotou/054/shiryo/attach/1361207.htm Accessed October 8, 2025.

Ministry of Education, Culture, Sports, Science and Technology (2018). Results of the 2018 survey on special needs education. Retrieved from https://www.mext.go.jp/a_menu/shotou/tokubetu/1402845_00003.htm. Accessed October 8, 2025.

Ministry of Health, Labour and Welfare (2021). Report on the survey on regional support functions and operation of support centers for people with developmental disabilities. Retrieved from https://www.mhlw.go.jp/stf/seisakunitsuite/bunya/0000194160_00015.html. Accessed October 8, 2025.

Miyaki, K., Song, Y., Htun, N. C., Tsutsumi, A., Hashimoto, H., Kawakami, N., ... Shimbo, T. (2012). Folate intake and depressive symptoms in Japanese workers considering SES and job stress factors: J-hope study. *BMC Psychiatry*, *12*, 33. https://doi.org/10.1186/1471-244X-12-33.

Mofokeng, M., & van der Wath, A. E. (2017). Challenges experienced by parents living with a child with attention deficit hyperactivity disorder. *Journal of Child and Adolescent Mental Health*, *29*(2), 137–145. https://doi.org/10.2989/17280583.2017.1364253.

Rimestad, M. L., Lambek, R., Zacher Christiansen, H., & Hougaard, E. (2019). Short- and long-term effects of parent training for preschool children with or at risk of ADHD: A systematic review and meta-analysis. *Journal of Attention Disorders*, *23*(5), 423–434. https://doi.org/10.1177/1087054716648775.

Saito, K., & Iida, J. (2022). *Guidelines for the diagnosis and treatment of attention deficit/hyperactivity disorder* (5th ed.). Jihou.

Shin, D. W., & Stein, M. A. (2008). Maternal depression predicts maternal use of corporal punishment in children with attention-deficit/hyperactivity disorder. *Yonsei Medical Journal*, *49*(4), 573–580. https://doi.org/10.3349/ymj.2008.49.4.573.

Song, H., Ito, R., & Watanabe, H. (2004). Research on support for children with high functioning autistic spectrum and Asperger's disorder and their families: relations between support and stress of parents. *The Japanese Journal of Autistic Spectrum*, *3*(1), 11–22.

Takaki, J., & Hibino, Y. (2014). Family-related opinions and stressful situations associated with psychological distress in women undergoing infertility treatment. *International Journal of Environmental Research and Public Health*, *11*(9), 9068–9081. https://doi.org/10.3390/ijerph110909068.

Takehara, K., Suto, M., & Kato, T. (2020). Parental psychological distress in the postnatal period in Japan: A population-based analysis of a national cross-sectional survey. *Scientific Reports*, *10*(1), 13770. https://doi.org/10.1038/s41598-020-70727-2.

Tani, I., Okada, R., Ohnishi, M., Nakajima, S., & Tsujii, M. (2010). Japanese version of home form of the ADHD-RS: An evaluation of its reliability and validity. *Research in Developmental Disabilities*, *31*(6), 1426–1433. https://doi.org/10.1016/j.ridd.2010.06.016.

Teraoka, A., Nakati, M., & Ninomiya, K. (2014). Acceptance of disability by a mother of a child with ADHD and social support. *International Nursing Care Research*, *13*(1), 25–32.

Theule, J., Wiener, J., Rogers, M. A., & Marton, I. (2011). Predicting parenting stress in families of children with ADHD: Parent and contextual factors. *Journal of Child and Family Studies*, *20*(5), 640–647. https://doi.org/10.1007/s10826-010-9439-7.

Theule, J., Wiener, J., Tannock, R., & Jenkins, J. M. (2013). Parenting stress in families of children with ADHD: A meta-analysis. *Journal of Emotional and Behavioral Disorders*, *21*(1), 3–17. https://doi.org/10.1177/1063426610387433.

Vaux, A. (1988). *Social support: Theory, research, and intervention.* Praeger.
von Elm, E., Altman, D. G., Egger, M., Pocock, S. J., Gøtzsche, P. C., Vandenbroucke, J. P., & STROBE Initiative (2007). The strengthening the reporting of observational studies in epidemiology (STROBE) statement: Guidelines for reporting observational studies. *Lancet*, *370*(9596), 1453–1457. https://doi.org/10.1016/S0140-6736(07)61602-X.
Walls, M., Cabral, H., Feinberg, E., & Silverstein, M. (2018). Association between changes in caregiver depressive symptoms and child attention-deficit/hyperactivity disorder symptoms. *Journal of Developmental and Behavioral Pediatrics*, *39*(5), 387–394. https://doi.org/10.1097/DBP.0000000000000562.
Watanabe, H., Ito, R., & Song, H. (2006). Stress on parents having children with high-functioning pervasive developmental disorder at transitional stages. *Japanese Journal of Developmental Disabilities*, *28*(1), 72–85.
Yamane, T. (2013). Developmental disorder parenting stressor index (DDPSI): Reliability and validity. *Shinrigaku Kenkyu: The Japanese Journal of Psychology*, *83*(6), 556–565. https://doi.org/10.4992/jjpsy.83.556.